DBMS-Mini Project
Team 4
SRN-PES2UG24CS088, PES2UG24CS095
Name-Aryan Burman,Atharva Singh
Problem Statement-Campus Task Board

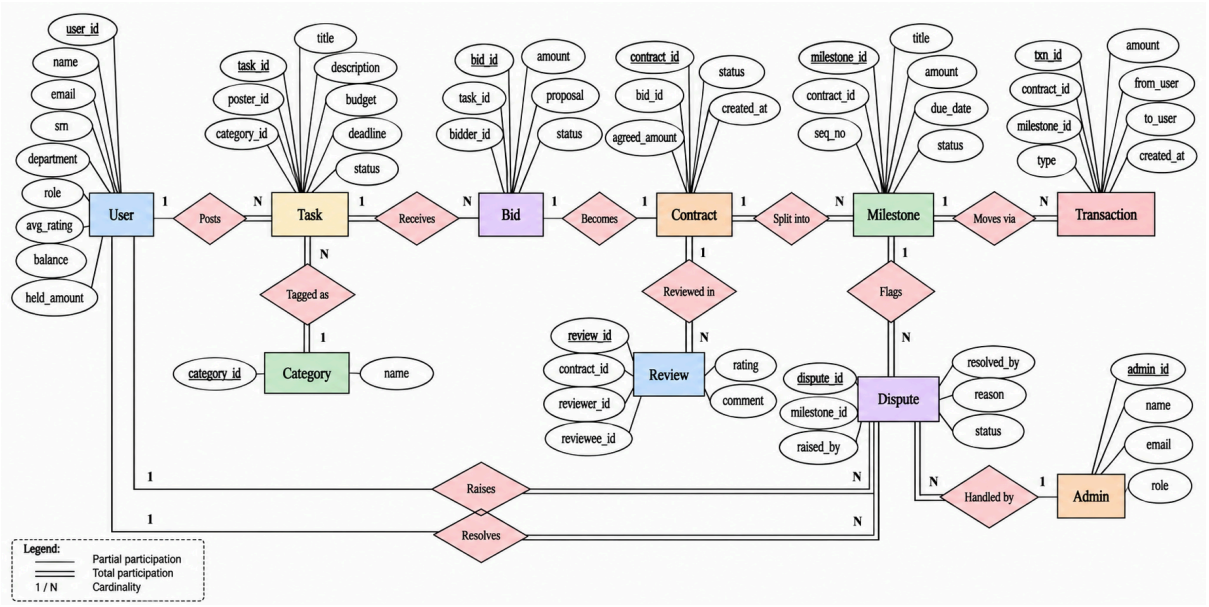
About the project

A website where students at your college post small paid tasks (poster design, coding help, video editing, etc.), other students bid on them, and payment is released in stages/steps as work is approved — instead of all upfront or all at the end. If someone doesn't deliver, an admin can step in and refund.

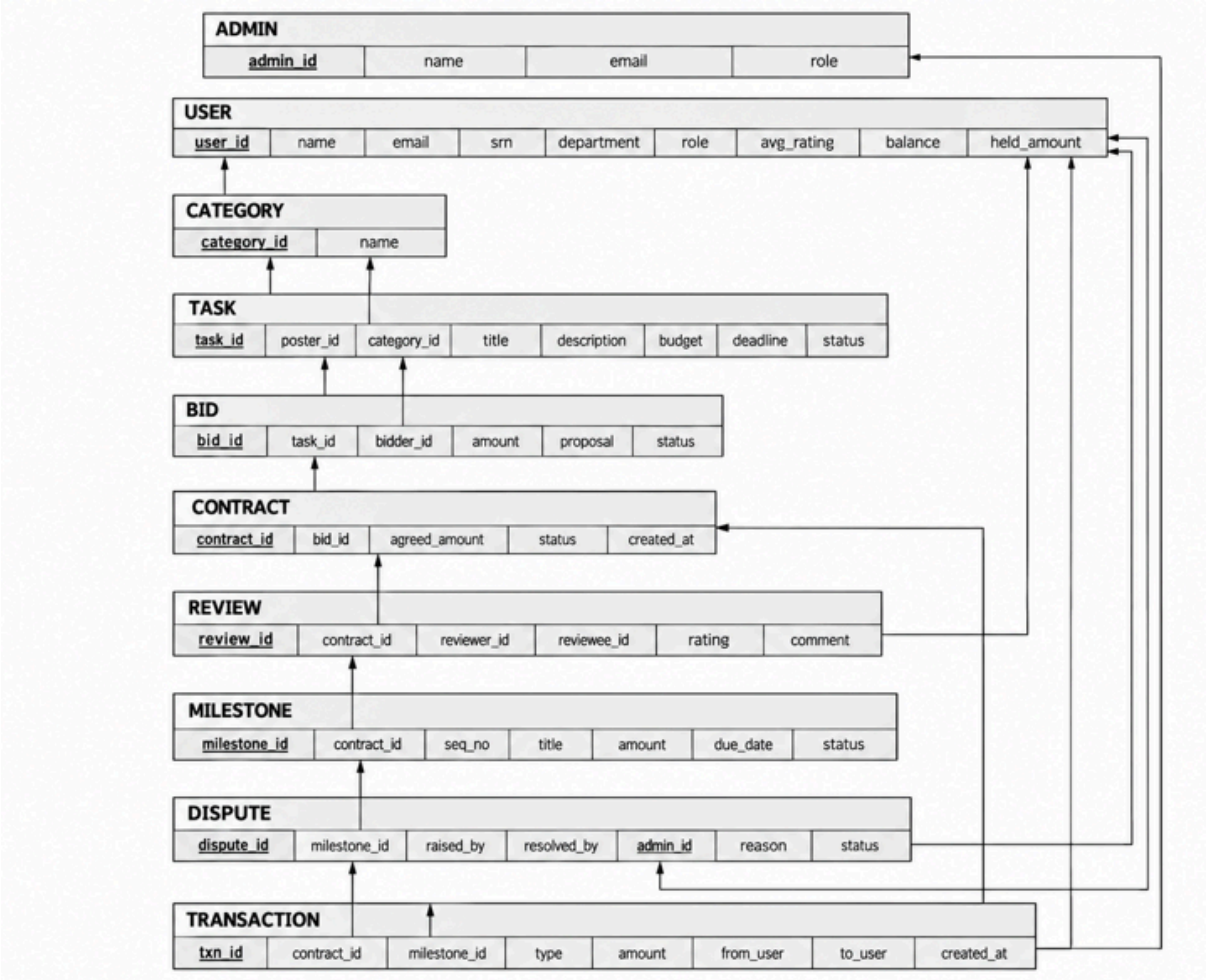
How it works:

1.Student posts a task with budget + deadline
2.Other students bid
3.Poster picks one; payment gets "held" (simulated, not real money)
4.Task is split into steps (e.g., draft → final); each step's payment releases only after approval
5.If something goes wrong, a flag/refund process kicks in
6.Both sides rate each other after

ER Diagram


user_id
name
email
srn
department
role
avg_rating
balance
held_amount
User
Posts
Task
title
description
budget
deadline
status
task_id
poster_id
category_id
Receives
Bid
bid_id
amount
proposal
status
task_id
bidder_id
Becomes
Contract
contract_id
status
created_at
bid_id
agreed_amount
Split into
Milestone
milestone_id
title
amount
due_date
status
contract_id
seq_no
Moves via
Transaction
txn_id
amount
from_user
to_user
created_at
contract_id
milestone_id
type
Tagged as
Category
category_id
name
Reviewed in
Review
review_id
contract_id
reviewer_id
reviewee_id
rating
comment
Flags
Dispute
dispute_id
milestone_id
raised_by
resolved_by
reason
status
Handled by
Admin
admin_id
name
email
role
Raises
Resolves
Legend:
Partial participation
Total participation
1 / N
Cardinality

Relational Schema



DDL and DML commands

DDL commands

```
mysql> SHOW TABLES;
+------------------------------+
| Tables_in_campus_task_board |
+------------------------------+
| admin                        |
| bid                          |
| category                     |
| contract                     |
| dispute                      |
| milestone                    |
| review                       |
| task                         |
| transaction_log              |
| users                        |
+------------------------------+
10 rows in set (0.03 sec)
```

```
mysql> CREATE TABLE USERS (
    ->     user_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     name         VARCHAR(100) NOT NULL,
    ->     email        VARCHAR(100) NOT NULL UNIQUE,
    ->     srn          VARCHAR(20)  NOT NULL UNIQUE,
    ->     department   VARCHAR(50),
    ->     role         ENUM('STUDENT','ADMIN') NOT NULL DEFAULT 'STUDENT',
    ->     avg_rating   DECIMAL(3,2)  NOT NULL DEFAULT 0,        -- derived, updated from REVIEW
    ->     balance      DECIMAL(10,2) NOT NULL DEFAULT 0 CHECK (balance >= 0),
    ->     held_amount  DECIMAL(10,2) NOT NULL DEFAULT 0 CHECK (held_amount >= 0)  -- money in escrow
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> DESC USERS;
+-------------+-------------------------+------+-----+---------+----------------+
| Field       | Type                    | Null | Key | Default | Extra          |
+-------------+-------------------------+------+-----+---------+----------------+
| user_id     | int                     | NO   | PRI | NULL    | auto_increment |
| name        | varchar(100)            | NO   |     | NULL    |                |
| email       | varchar(100)            | NO   | UNI | NULL    |                |
| srn         | varchar(20)             | NO   | UNI | NULL    |                |
| department  | varchar(50)             | YES  |     | NULL    |                |
| role        | enum('STUDENT','ADMIN') | NO   |     | STUDENT |                |
| avg_rating  | decimal(3,2)            | NO   |     | 0.00    |                |
| balance     | decimal(10,2)           | NO   |     | 0.00    |                |
| held_amount | decimal(10,2)           | NO   |     | 0.00    |                |
+-------------+-------------------------+------+-----+---------+----------------+
9 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE ADMIN (
    ->     admin_id     INT AUTO_INCREMENT PRIMARY KEY,
    ->     name         VARCHAR(100) NOT NULL,
    ->     email        VARCHAR(100) NOT NULL UNIQUE,
    ->     role         VARCHAR(50) DEFAULT 'MODERATOR'
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC ADMIN;
+----------+--------------+------+-----+-----------+----------------+
| Field    | Type         | Null | Key | Default   | Extra          |
+----------+--------------+------+-----+-----------+----------------+
| admin_id | int          | NO   | PRI | NULL      | auto_increment |
| name     | varchar(100) | NO   |     | NULL      |                |
| email    | varchar(100) | NO   | UNI | NULL      |                |
| role     | varchar(50)  | YES  |     | MODERATOR |                |
+----------+--------------+------+-----+-----------+----------------+
4 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE CATEGORY (
    ->     category_id  INT AUTO_INCREMENT PRIMARY KEY,
    ->     name         VARCHAR(50) NOT NULL UNIQUE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC CATEGORY;
+-------------+-------------+------+-----+---------+----------------+
| Field       | Type        | Null | Key | Default | Extra          |
+-------------+-------------+------+-----+---------+----------------+
| category_id | int         | NO   | PRI | NULL    | auto_increment |
| name        | varchar(50) | NO   | UNI | NULL    |                |
+-------------+-------------+------+-----+---------+----------------+
2 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE TASK (
    ->     task_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     poster_id    INT NOT NULL,
    ->     category_id  INT,
    ->     title        VARCHAR(150) NOT NULL,
    ->     description  TEXT,
    ->     budget       DECIMAL(10,2) NOT NULL CHECK (budget > 0),
    ->     deadline     DATE,
    ->     status       ENUM('OPEN','ASSIGNED','COMPLETED','CANCELLED') NOT NULL DEFAULT 'OPEN',
    ->     FOREIGN KEY (poster_id)   REFERENCES USERS(user_id),
    ->     FOREIGN KEY (category_id) REFERENCES CATEGORY(category_id)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> DESC TASK;
+-------------+---------------------------------------------+------+-----+---------+----------------+
| Field       | Type                                        | Null | Key | Default | Extra          |
+-------------+---------------------------------------------+------+-----+---------+----------------+
| task_id     | int                                         | NO   | PRI | NULL    | auto_increment |
| poster_id   | int                                         | NO   | MUL | NULL    |                |
| category_id | int                                         | YES  | MUL | NULL    |                |
| title       | varchar(150)                                | NO   |     | NULL    |                |
| description | text                                        | YES  |     | NULL    |                |
| budget      | decimal(10,2)                               | NO   |     | NULL    |                |
| deadline    | date                                        | YES  |     | NULL    |                |
| status      | enum('OPEN','ASSIGNED','COMPLETED','CANCELLED') | NO   |     | OPEN    |                |
+-------------+---------------------------------------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE CONTRACT (
    ->     contract_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     bid_id         INT NOT NULL UNIQUE,   -- UNIQUE turns this 1:N FK into a 1:1
    ->     agreed_amount  DECIMAL(10,2) NOT NULL,
    ->     status         ENUM('ACTIVE','COMPLETED','DISPUTED','REFUNDED') NOT NULL DEFAULT 'ACTIVE',
    ->     created_at     TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (bid_id) REFERENCES BID(bid_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC CONTRACT;
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                                          | Null | Key | Default           | Extra             |
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| contract_id  | int                                           | NO   | PRI | NULL              | auto_increment    |
| bid_id       | int                                           | NO   | UNI | NULL              |                   |
| agreed_amount | decimal(10,2)                                | NO   |     | NULL              |                   |
| status       | enum('ACTIVE','COMPLETED','DISPUTED','REFUNDED') | NO   |     | ACTIVE            |                   |
| created_at   | timestamp                                     | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+-----------------------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE REVIEW (
    ->     review_id     INT AUTO_INCREMENT PRIMARY KEY,
    ->     contract_id   INT NOT NULL,
    ->     reviewer_id   INT NOT NULL,   -- FK -> USERS, person giving the review
    ->     reviewee_id   INT NOT NULL,   -- FK -> USERS, person being reviewed
    ->     rating        INT NOT NULL CHECK (rating BETWEEN 1 AND 5),
    ->     comment       TEXT,
    ->     FOREIGN KEY (contract_id) REFERENCES CONTRACT(contract_id),
    ->     FOREIGN KEY (reviewer_id) REFERENCES USERS(user_id),
    ->     FOREIGN KEY (reviewee_id) REFERENCES USERS(user_id),
    ->     UNIQUE (contract_id, reviewer_id)   -- one review per person per contract
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> DESC REVIEW;
+-------------+------+------+-----+---------+----------------+
| Field       | Type | Null | Key | Default | Extra          |
+-------------+------+------+-----+---------+----------------+
| review_id   | int  | NO   | PRI | NULL    | auto_increment |
| contract_id | int  | NO   | MUL | NULL    |                |
| reviewer_id | int  | NO   | MUL | NULL    |                |
| reviewee_id | int  | NO   | MUL | NULL    |                |
| rating      | int  | NO   |     | NULL    |                |
| comment     | text | YES  |     | NULL    |                |
+-------------+------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE MILESTONE (
    ->     milestone_id  INT AUTO_INCREMENT PRIMARY KEY,   -- surrogate key, simpler than a composite
    ->     contract_id   INT NOT NULL,
    ->     seq_no        INT NOT NULL,
    ->     title         VARCHAR(100) NOT NULL,
    ->     amount        DECIMAL(10,2) NOT NULL CHECK (amount > 0),
    ->     due_date      DATE,
    ->     status        ENUM('PENDING','SUBMITTED','APPROVED','FLAGGED') NOT NULL DEFAULT 'PENDING',
    ->     FOREIGN KEY (contract_id) REFERENCES CONTRACT(contract_id),
    ->     UNIQUE (contract_id, seq_no)   -- step 1, 2, 3... within one contract
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESC MILESTONE;
+-------------+-----------------------------------------------+------+-----+---------+----------------+
| Field       | Type                                          | Null | Key | Default | Extra          |
+-------------+-----------------------------------------------+------+-----+---------+----------------+
| milestone_id | int                                          | NO   | PRI | NULL    | auto_increment |
| contract_id | int                                           | NO   | MUL | NULL    |                |
| seq_no      | int                                           | NO   |     | NULL    |                |
| title       | varchar(100)                                  | NO   |     | NULL    |                |
| amount      | decimal(10,2)                                 | NO   |     | NULL    |                |
| due_date    | date                                          | YES  |     | NULL    |                |
| status      | enum('PENDING','SUBMITTED','APPROVED','FLAGGED') | NO   |     | PENDING |                |
+-------------+-----------------------------------------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE DISPUTE (
    ->     dispute_id    INT AUTO_INCREMENT PRIMARY KEY,
    ->     milestone_id  INT NOT NULL,
    ->     raised_by     INT NOT NULL,   -- FK -> USERS, student who filed it
    ->     resolved_by   INT,            -- FK -> USERS, nullable until settled
    ->     admin_id      INT,            -- FK -> ADMIN, who actually handled it
    ->     reason        TEXT NOT NULL,
    ->     status        ENUM('OPEN','REFUNDED','REJECTED') NOT NULL DEFAULT 'OPEN',
    ->     FOREIGN KEY (milestone_id) REFERENCES MILESTONE(milestone_id),
    ->     FOREIGN KEY (raised_by)    REFERENCES USERS(user_id),
    ->     FOREIGN KEY (resolved_by)  REFERENCES USERS(user_id),
    ->     FOREIGN KEY (admin_id)     REFERENCES ADMIN(admin_id)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> DESC DISPUTE;
+--------------+-----------------------------------+------+-----+---------+----------------+
| Field        | Type                              | Null | Key | Default | Extra          |
+--------------+-----------------------------------+------+-----+---------+----------------+
| dispute_id   | int                               | NO   | PRI | NULL    | auto_increment |
| milestone_id | int                               | NO   | MUL | NULL    |                |
| raised_by    | int                               | NO   | MUL | NULL    |                |
| resolved_by  | int                               | YES  | MUL | NULL    |                |
| admin_id     | int                               | YES  | MUL | NULL    |                |
| reason       | text                              | NO   |     | NULL    |                |
| status       | enum('OPEN','REFUNDED','REJECTED') | NO   |     | OPEN    |                |
+--------------+-----------------------------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

```
mysql>
mysql> CREATE TABLE TRANSACTION_LOG (
    ->     txn_id        INT AUTO_INCREMENT PRIMARY KEY,
    ->     contract_id   INT NOT NULL,
    ->     milestone_id  INT,   -- nullable: the initial HOLD has no milestone yet
    ->     type          ENUM('HOLD','RELEASE','REFUND') NOT NULL,
    ->     amount        DECIMAL(10,2) NOT NULL CHECK (amount > 0),
    ->     from_user     INT NOT NULL,
    ->     to_user       INT NOT NULL,
    ->     created_at    TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     FOREIGN KEY (contract_id)  REFERENCES CONTRACT(contract_id),
    ->     FOREIGN KEY (milestone_id) REFERENCES MILESTONE(milestone_id),
    ->     FOREIGN KEY (from_user)    REFERENCES USERS(user_id),
    ->     FOREIGN KEY (to_user)      REFERENCES USERS(user_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESC TRANSACTION_LOG;
+--------------+---------------------------------+------+-----+-------------------+-------------------+
| Field        | Type                            | Null | Key | Default           | Extra             |
+--------------+---------------------------------+------+-----+-------------------+-------------------+
| txn_id       | int                             | NO   | PRI | NULL              | auto_increment    |
| contract_id  | int                             | NO   | MUL | NULL              |                   |
| milestone_id | int                             | YES  | MUL | NULL              |                   |
| type         | enum('HOLD','RELEASE','REFUND') | NO   |     | NULL              |                   |
| amount       | decimal(10,2)                   | NO   |     | NULL              |                   |
| from_user    | int                             | NO   | MUL | NULL              |                   |
| to_user      | int                             | NO   | MUL | NULL              |                   |
| created_at   | timestamp                       | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+---------------------------------+------+-----+-------------------+-------------------+
8 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE BID (
    ->     bid_id       INT AUTO_INCREMENT PRIMARY KEY,
    ->     task_id      INT NOT NULL,
    ->     bidder_id    INT NOT NULL,
    ->     amount       DECIMAL(10,2) NOT NULL CHECK (amount > 0),
    ->     proposal     TEXT,
    ->     status       ENUM('PENDING','ACCEPTED','REJECTED') NOT NULL DEFAULT 'PENDING',
    ->     FOREIGN KEY (task_id)   REFERENCES TASK(task_id),
    ->     FOREIGN KEY (bidder_id) REFERENCES USERS(user_id),
    ->     UNIQUE (task_id, bidder_id)   -- one bid per student per task
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> DESC BID;
+-----------+---------------------------------------+------+-----+---------+----------------+
| Field     | Type                                  | Null | Key | Default | Extra          |
+-----------+---------------------------------------+------+-----+---------+----------------+
| bid_id    | int                                   | NO   | PRI | NULL    | auto_increment |
| task_id   | int                                   | NO   | MUL | NULL    |                |
| bidder_id | int                                   | NO   | MUL | NULL    |                |
| amount    | decimal(10,2)                         | NO   |     | NULL    |                |
| proposal  | text                                  | YES  |     | NULL    |                |
| status    | enum('PENDING','ACCEPTED','REJECTED') | NO   |     | PENDING |                |
+-----------+---------------------------------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

DML Commands

```
mysql> INSERT INTO USERS (name, email, srn, department, role, balance, held_amount) VALUES
    -> ('Aryan Sharma', 'aryan@pes.edu', 'PES2UG24CS088', 'CSE', 'STUDENT', 1000.00, 0.00),
    -> ('Diya Rao',     'diya@pes.edu',  'PES2UG24CS045', 'CSE', 'STUDENT',  500.00, 0.00),
    -> ('Kabir Mehta',  'kabir@pes.edu', 'PES2UG24EC012', 'ECE', 'STUDENT',  300.00, 0.00);
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO ADMIN (name, email, role) VALUES
    -> ('Prof. Nandini Iyer', 'nandini.admin@pes.edu', 'MODERATOR');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO CATEGORY (name) VALUES
    -> ('Poster Design'), ('Video Editing'), ('Coding Help'), ('Event Assistance');
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO TASK (poster_id, category_id, title, description, budget, deadline, status) VALUES
    -> (1, 1, 'Cultural fest poster', 'A3 poster for the annual cultural fest', 500.00, '2026-10-05', 'ASSIGNED'),
    -> (2, 3, 'Debug a Flask app',    'Fix a login bug in a small Flask project', 300.00, '2026-10-02', 'OPEN');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO BID (task_id, bidder_id, amount, proposal, status) VALUES
    -> (1, 2, 450.00, 'I can start today, 2 revisions included', 'ACCEPTED'),
    -> (1, 3, 480.00, 'I have poster design experience',          'REJECTED'),
    -> (2, 3, 280.00, 'I know Flask auth well',                   'PENDING');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- bid 1 (Diya's bid on Aryan's poster task) was accepted -> becomes a contract
mysql> INSERT INTO CONTRACT (bid_id, agreed_amount, status) VALUES
    -> (1, 450.00, 'ACTIVE');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO MILESTONE (contract_id, seq_no, title, amount, due_date, status) VALUES
    -> (1, 1, 'Draft poster', 200.00, '2026-10-02', 'PENDING'),
    -> (1, 2, 'Final poster', 250.00, '2026-10-05', 'PENDING');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> -- bids on a task, cheapest first
mysql> SELECT b.bid_id, u.name AS bidder, b.amount, b.status
    -> FROM BID b JOIN USERS u ON b.bidder_id = u.user_id
    -> WHERE b.task_id = 1
    -> ORDER BY b.amount;
+--------+-------------+--------+----------+
| bid_id | bidder      | amount | status   |
+--------+-------------+--------+----------+
|      1 | Diya Rao    | 450.00 | ACCEPTED |
|      2 | Kabir Mehta | 480.00 | REJECTED |
+--------+-------------+--------+----------+
2 rows in set (0.00 sec)

mysql>
mysql> -- money currently sitting in escrow, per user
mysql> SELECT user_id, name, held_amount FROM USERS WHERE held_amount > 0;
Empty set (0.00 sec)

mysql>
mysql> -- average rating per user, computed live from REVIEW (not the stored avg_rating column)
mysql> SELECT u.user_id, u.name, ROUND(AVG(r.rating), 2) AS computed_avg_rating
    -> FROM USERS u JOIN REVIEW r ON u.user_id = r.reviewee_id
    -> GROUP BY u.user_id, u.name;
Empty set (0.01 sec)

mysql>
mysql> -- milestones for one contract, in delivery order
mysql> SELECT seq_no, title, amount, status FROM MILESTONE WHERE contract_id = 1 ORDER BY seq_no;
+--------+--------------+--------+---------+
| seq_no | title        | amount | status  |
+--------+--------------+--------+---------+
|      1 | Draft poster | 200.00 | PENDING |
|      2 | Final poster | 250.00 | PENDING |
+--------+--------------+--------+---------+
2 rows in set (0.00 sec)

mysql>
mysql> -- disputes each admin has closed
mysql> SELECT d.dispute_id, d.reason, d.status, a.name AS handled_by
    -> FROM DISPUTE d JOIN ADMIN a ON d.admin_id = a.admin_id;
Empty set (0.00 sec)
```

```
mysql> -- accept-bid transaction: moves agreed_amount from poster's balance into
mysql> -- held_amount, accepts the winning bid, rejects the rest, marks task ASSIGNED
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>   -- lock the poster's row so a concurrent accept can't double-spend the same balance
mysql>   SELECT balance INTO @poster_balance FROM USERS WHERE user_id = 1 FOR UPDATE;
Query OK, 1 row affected (0.00 sec)

mysql>
mysql>   UPDATE USERS
    ->   SET balance = balance - 450.00, held_amount = held_amount + 450.00
    ->   WHERE user_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>   UPDATE BID SET status = 'ACCEPTED' WHERE bid_id = 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>   UPDATE BID SET status = 'REJECTED' WHERE task_id = 1 AND bid_id <> 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>   UPDATE TASK SET status = 'ASSIGNED' WHERE task_id = 1;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql>   INSERT INTO TRANSACTION_LOG (contract_id, milestone_id, type, amount, from_user, to_user)
    ->   VALUES (1, NULL, 'HOLD', 450.00, 1, 2);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>
mysql> -- approve-milestone transaction: releases escrow from poster to worker
mysql> START TRANSACTION;
```

```
mysql>
mysql>   UPDATE USERS SET held_amount = held_amount - 200.00 WHERE user_id = 1;   -- poster
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>   UPDATE USERS SET balance     = balance     + 200.00 WHERE user_id = 2;   -- worker
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>   UPDATE MILESTONE SET status = 'APPROVED' WHERE milestone_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>   INSERT INTO TRANSACTION_LOG (contract_id, milestone_id, type, amount, from_user, to_user)
    ->   VALUES (1, 1, 'RELEASE', 200.00, 1, 2);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>
mysql> -- dispute raised on the second milestone, then resolved by an admin as a refund
mysql> INSERT INTO DISPUTE (milestone_id, raised_by, reason, status)
    -> VALUES (2, 1, 'Final poster was never delivered', 'OPEN');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql>   UPDATE USERS SET held_amount = held_amount - 250.00 WHERE user_id = 1;   -- release the hold
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>   UPDATE USERS SET balance     = balance     + 250.00 WHERE user_id = 1;   -- refund back to poster
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>   UPDATE MILESTONE SET status = 'FLAGGED' WHERE milestone_id = 2;
Query OK, 1 row affected (0.00 sec)
```

```
mysql>
mysql>   UPDATE DISPUTE
    ->   SET status = 'REFUNDED', resolved_by = 1, admin_id = 1
    ->   WHERE dispute_id = 1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>   INSERT INTO TRANSACTION_LOG (contract_id, milestone_id, type, amount, from_user, to_user)
    ->   VALUES (1, 2, 'REFUND', 250.00, 2, 1);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```